Department of Applied Mathematics and Statistics
COLORADO SCHOOL OF MINES
MATH 455: Partial Differential Equations

Assignment #5 - Laplace's Equation
Due Thursday, November 5, 2026

1. (14 points) Let $a, b > 0$ and $g \in \mathrm{PC}^1[0, a]$ be given. Find the solution $u(x, y)$ of

$$\Delta u = 0$$

on the rectangle $\Omega = [0, a] \times [0, b]$, with

$$\frac{\partial u}{\partial x}(0, y) = \frac{\partial u}{\partial x}(a, y) = 0$$

for $0 < y < b$ and

$$\frac{\partial u}{\partial y}(x, 0) = 0 \qquad \text{and} \qquad u(x, b) = g(x)$$

for $0 < x < a$.
Hint: It might be easiest to use sinh and cosh in some places rather than real exponentials.

Solution —

$$\Delta u = 0 \implies \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

Since this is a linear PDE, we proceed with separation of variables, assuming the solution takes the (separated) form $u(x, y) = X(x)Y(y)$

$$\implies X''(x)Y(y) = -X(x)Y''(y)$$
$$\implies \frac{X''(x)}{X(x)} = -\frac{Y''(y)}{Y(y)} = \lambda$$

This equation takes the form $F(x) = G(y)$ for all $x, y \implies$ both must be equal to some constant λ (fix $x = x_0$, and then $F(x_0) = G(y)$ for all $y \implies G(y)$ is constant $\implies F(x)$ is constant since they're equal to each other). This yields the two ordinary differential equations

$$\begin{cases} X'' - \lambda X = 0 \\ Y'' + \lambda Y = 0 \end{cases}$$

whose solutions depend on the sign of λ.

For $\lambda < 0$:

$$X(x) = c_0 \cos\left(\sqrt{-\lambda}x\right) + c_1 \sin\left(\sqrt{-\lambda}x\right) \qquad Y(y) = c_2 e^{\sqrt{\lambda}y} + c_3 e^{-\sqrt{\lambda}y}$$

For $\lambda = 0$:

$$X(x) = Ax + B \qquad Y(y) = Cy + D$$

For $\lambda > 0$:

$$X(x) = c_4 e^{\sqrt{\lambda}x} + c_5 e^{-\sqrt{\lambda}x} \qquad Y(y) = c_6 \cos\left(\sqrt{\lambda}y\right) + c_7 \sin\left(\sqrt{\lambda}y\right)$$

2. (16 points) Let $a > 0$ be given and find (using polar coordinates) the solution $u(r, \theta)$ of Laplace's equation

$$\Delta u = 0$$

on the wedge

$$\Omega = \left\{(r, \theta) : 0 \leq r < a, 0 \leq \theta \leq \frac{\pi}{3}\right\}$$

with

$$u(r, 0) = u\left(r, \frac{\pi}{3}\right) = 0$$

for $0 \leq r < a$ and

$$u(a, \theta) = a^2$$

for $0 \leq \theta \leq \frac{\pi}{3}$. In particular, make sure to only find solutions that remain bounded as $r \to 0$.

3. (20 points) Let $0 < R_1 < R_2$ be given and g be a PC^1 function. Find the single-valued solution of Laplace's equation on the annulus

$$\Omega = \{(r, \theta) : R_1 < r < R_2, -\pi \leq \theta < \pi\}$$

with the boundary conditions

$$u(R_1, \theta) = 0 \qquad \text{and} \qquad u(R_2, \theta) = g(\theta)$$

for $-\pi \leq \theta < \pi$. In particular, use the periodic boundary conditions

$$u(r, -\pi) = u(r, \pi) \qquad \text{and} \qquad \frac{\partial u}{\partial \theta}(r, -\pi) = \frac{\partial u}{\partial \theta}(r, \pi)$$

for $R_1 < r < R_2$ in order to compute the solution.